

JEE MAIN REVISION MASTER: LINEAR PROGRAMMING 2026 2 QUESTIONS 3.33% WEIGHTAGE

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 16 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Linear Programming - Core Application	16 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Linear Programming - Core Application (Very High Yield — 16 PYQs)

Core Principle: Key problem pattern in Kcet Linear Programming.

■■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "In Linear Programming Problem (LPP), the objective function $Z=ax+by$ has the same maximum value at two corner points. The number of points ..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "The corner points of the feasible region determined by the system of linear constraints are $(0,10),(5,5),(15,15),(0,20)$. Let $z=px+qy$ wh..." [Ans: D]
- [2025 Shift] Q#3 (MCQ): "Consider the following statements: Statement (I): In a LPP, the objective function is always linear. Statement (II): In a LPP, the linear ine..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

In Linear Programming Problem (LPP), the objective function $Z=ax+by$ has the same maximum value at two corner points. The number of points at which $Z_{\text{(max)}}$ occurs is

- (A) 1
- (B) 2
- (C) 0
- (D) Infinity

Correct Answer: None

Solution Analysis: If the objective function $Z=ax+by$ has the same maximum value at two corner points of the feasible region, then Z will have the same value at every point on the line segment joining those two corner points. This happens because $Z=ax+by$ is a linear function. So, if it is maximum at two endpoints of an edge of the feasible region, then it remains maximum at all points on that edge. Hence, $Z_{\text{(max)}}$ occurs at infinitely many points. So, the correct option is: Option D: Infinity